Data Analytics II

March 23, 2026

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[2]:
```
dataset = pd.read_csv('Social_Network_Ads.csv')
```

[3]:
```
X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values
```

[4]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)
```

[5]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

[6]:
```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state=0)Confusion Matrix:
[[65  3]
 [ 8 24]]


classifier.fit(X_train, y_train)
```

[6] : LogisticRegression(random_state=0)

[7] :
```
y_pred = classifier.predict(X_test)
```

[8] :
```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[65  3]
 [ 8 24]]
```

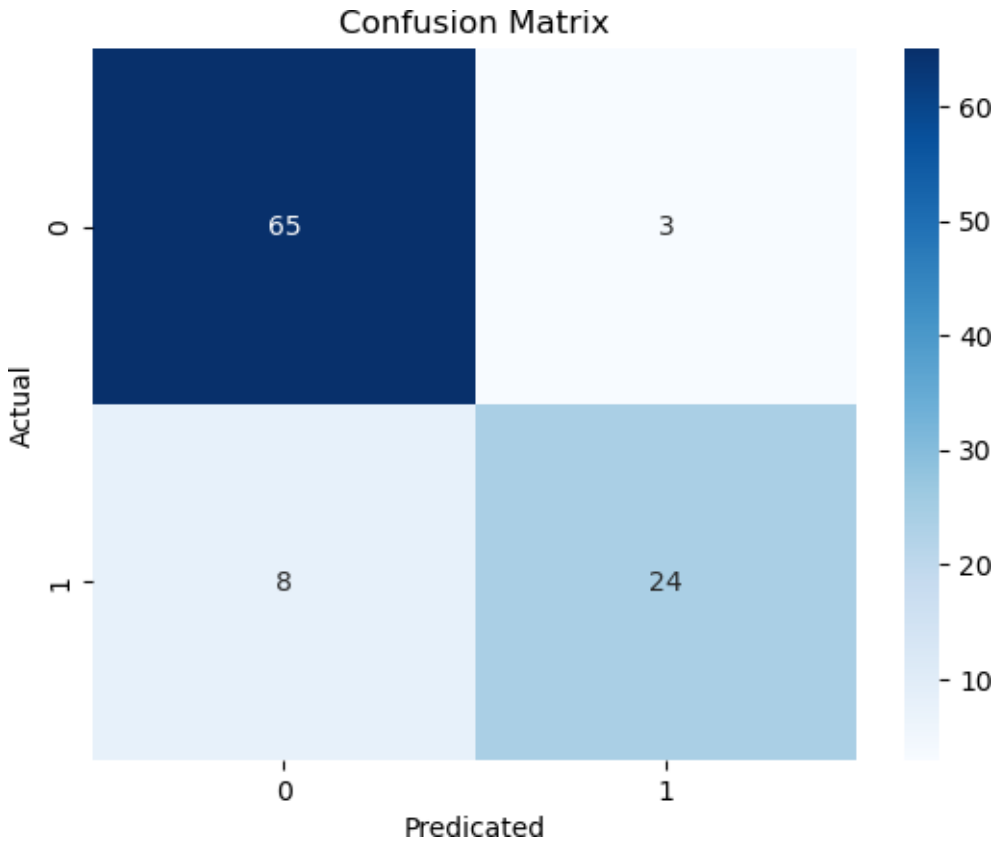
[12]:
```
TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

print("TP:", TP)
print("FP:", FP)
print("TN:", TN)
print("FN:", FN)
```

```
TP: 24
FP: 3
TN: 65
FN: 8
```

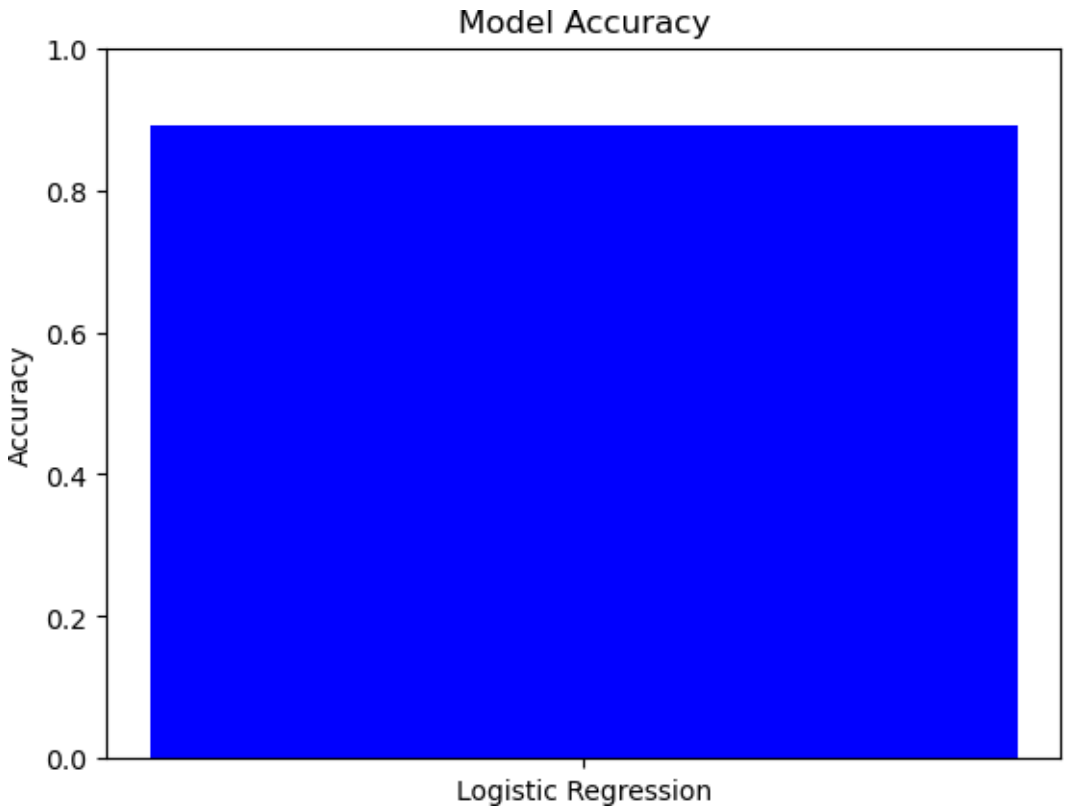
[13]:
```
accuracy = (TP + TN) / (TP + TN + FP + FN)
error_rate = 1 - accuracy
precision = TP / (TP + FP)
recall = TP / (TP + FN)

print("\nAccuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)
```

```
Accuracy: 0.89
Error Rate: 0.10999999999999999
Precision: 0.8888888888888888
Recall: 0.75
```

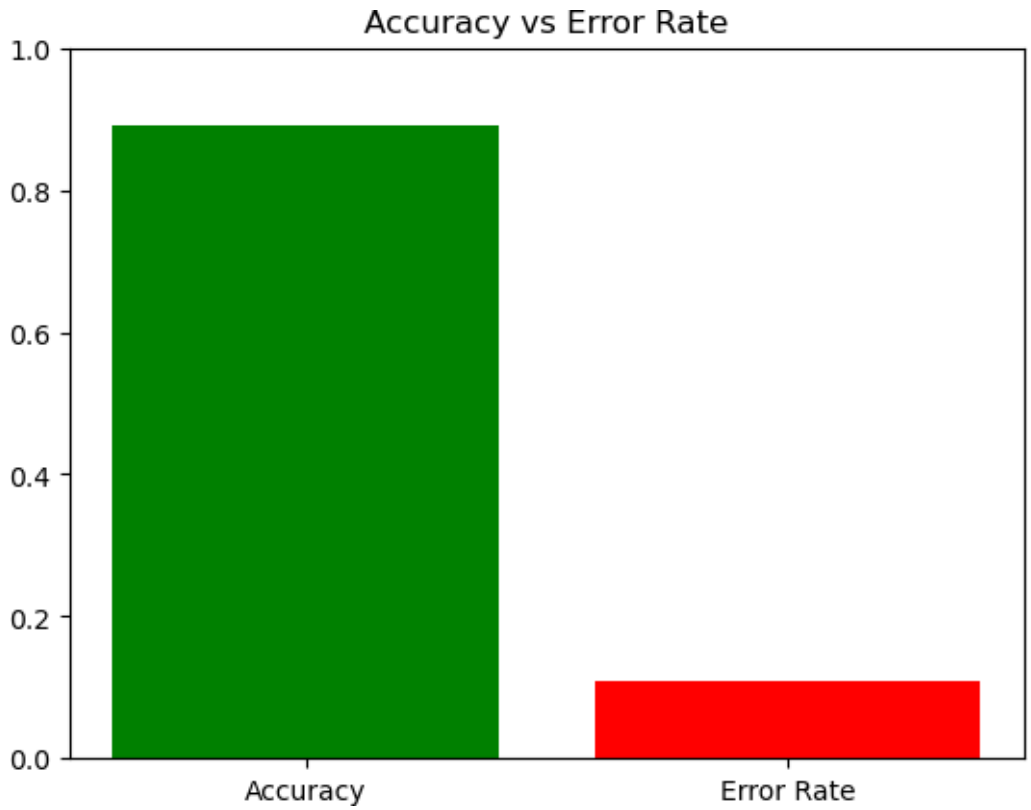
[24]:
```
import seaborn as sns
sns.heatmap(cm,annot=True,fmt='d',cmap='Blues')
plt.xlabel("Predicated")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



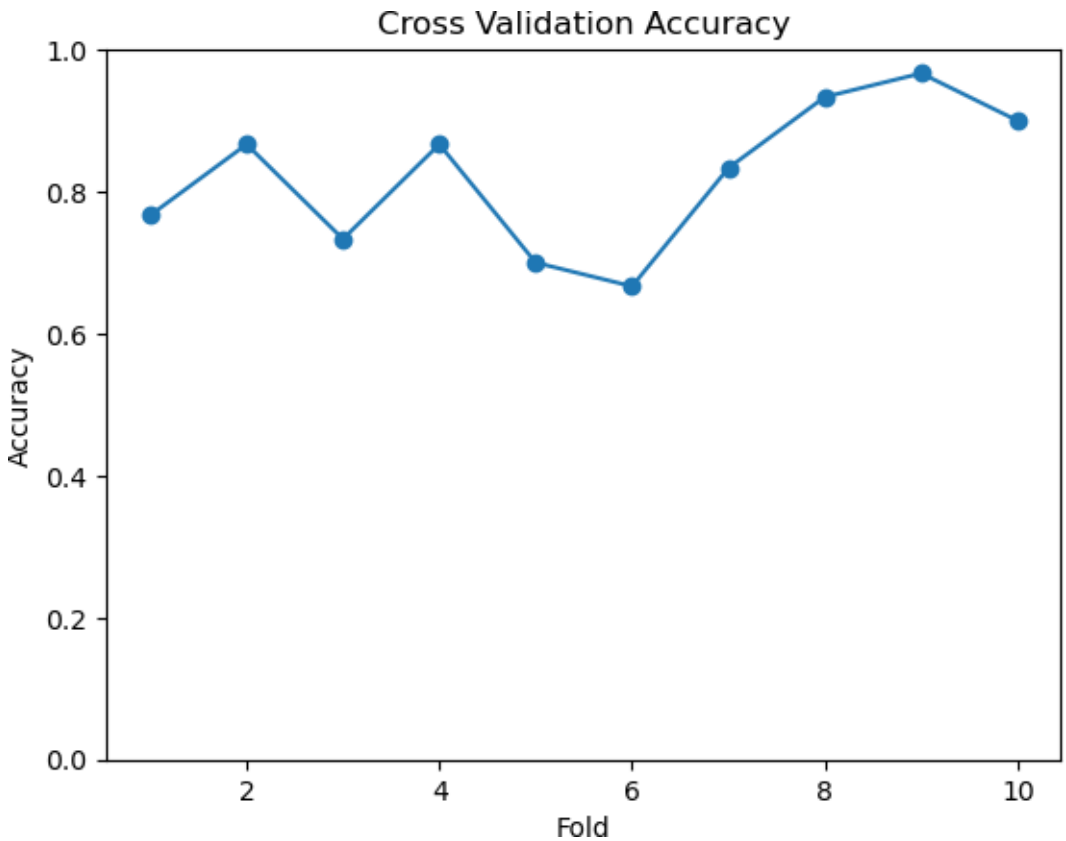
[25]:
```
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Plot
plt.bar(['Logistic Regression'], [accuracy], color='blue')
plt.ylim(0, 1)
plt.ylabel("Accuracy")
plt.title("Model Accuracy")
plt.show()
```



[26]:
```
error_rate = 1 - accuracy

labels = ['Accuracy', 'Error Rate']
values = [accuracy, error_rate]

plt.bar(labels, values, color=['green', 'red'])
plt.ylim(0, 1)
plt.title("Accuracy vs Error Rate")
plt.show()
```

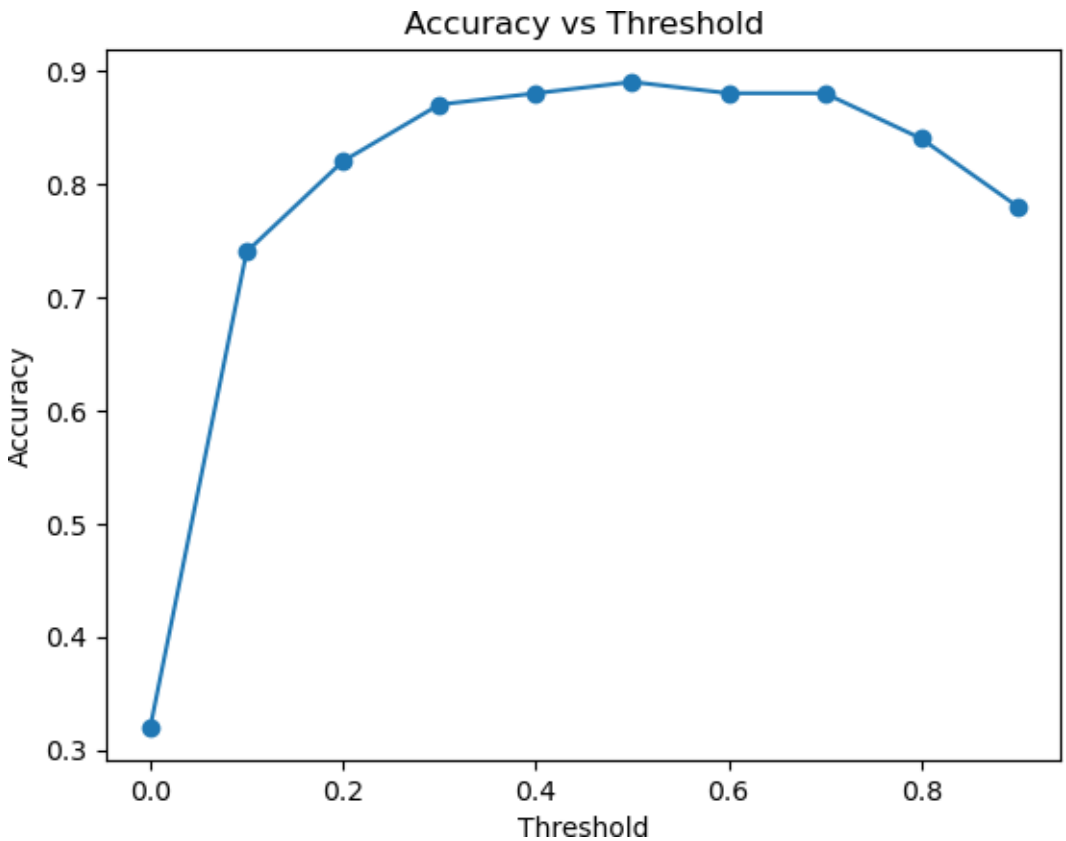


[27]:
```
from sklearn.model_selection import cross_val_score
import numpy as np

accuracies = cross_val_score(estimator=model, X=X_train, y=y_train, cv=10)

# Plot
plt.plot(range(1, 11), accuracies, marker='o')
plt.title("Cross Validation Accuracy")
plt.xlabel("Fold")
plt.ylabel("Accuracy")
plt.ylim(0, 1)
plt.show()

print("Mean Accuracy:", accuracies.mean())
```



Mean Accuracy: 0.8233333333333335

[29]:
```
import numpy as np

y_prob = model.predict_proba(X_test)[:, 1]

thresholds = np.arange(0.0, 1.0, 0.1)
acc_list = []

for t in thresholds:
    y_pred_thresh = (y_prob >= t).astype(int)
    acc_list.append(accuracy_score(y_test, y_pred_thresh))

plt.plot(thresholds, acc_list, marker='o')
plt.xlabel("Threshold")
plt.ylabel("Accuracy")
plt.title("Accuracy vs Threshold")
plt.show()
```


Accuracy vs Threshold
Accuracy
Threshold
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8

[ ]:

```
```